

eCommerce Web Analytics

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Introduction

E-commerce depends on product recommendations. However, relational databases store data in rows and struggle with relationships. While, graphs can model connections directly.

Graph Algorithms Used:

1. Page Rank
2. Louvain Community Detection
3. Node Similarity

Business Case Scenario

Imagine we work at Amazon:

Data: Amazon Sales Dataset from Kaggle, 1K+ Amazon sales records with product Ratings and reviews listed on amazon.com

Key idea: Use customer behavior to connect products.

Goal: Use the data to improve:

- Product recommendations
- Cross-sell opportunities
- Bundle discovery

Why Not Relational Databases?

1. Expensive joins
2. Poor relationship visibility
3. Difficult to scale for connections

Graph Design

Nodes:

- Users
- Products

Relationships:

- REVIEWED (Users to Products)
- CO_REVIEWED (Between Products reviewed by same Users)

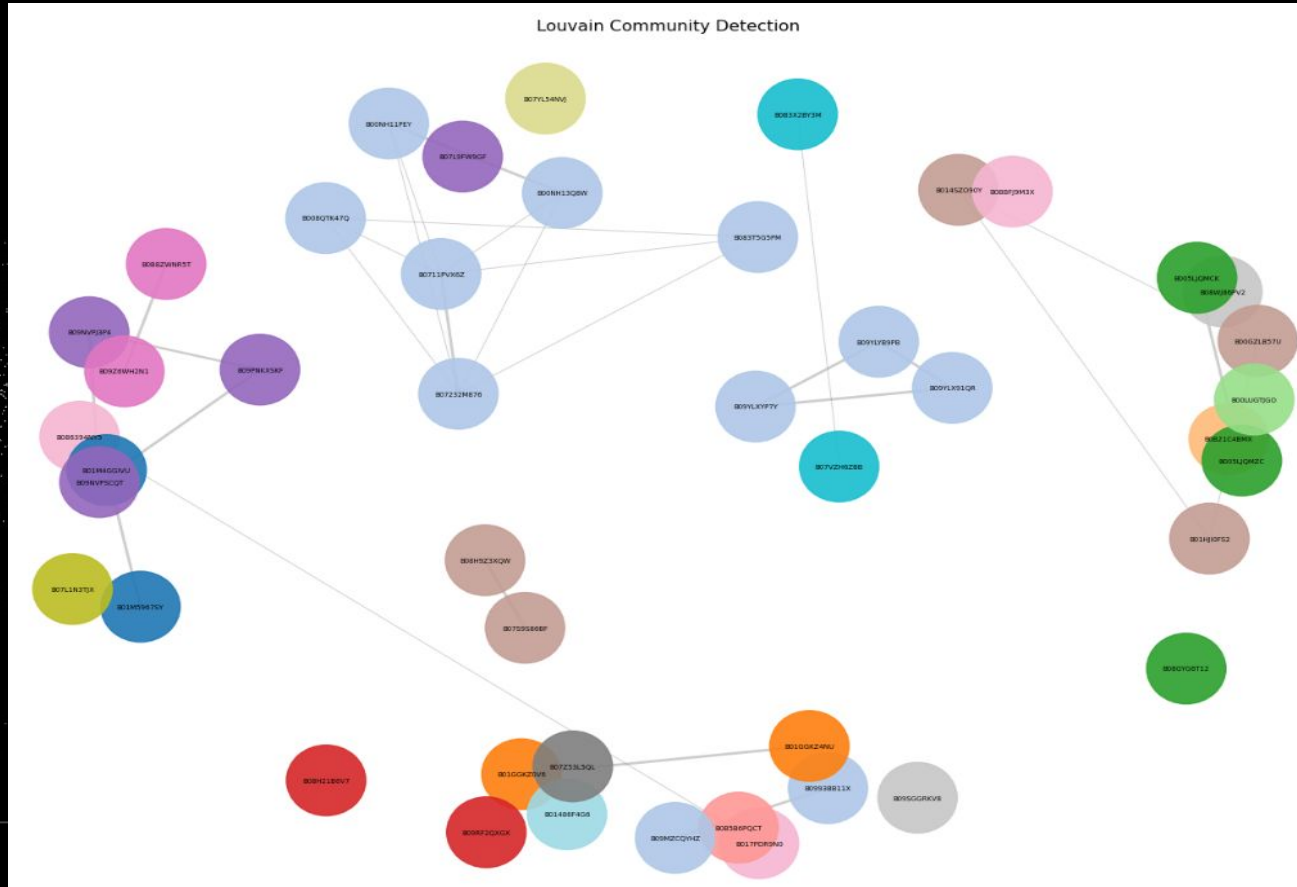
Edge Meaning: Shared Reviewers

Edge Weight: Strength of Connection (How often products were reviewed together)

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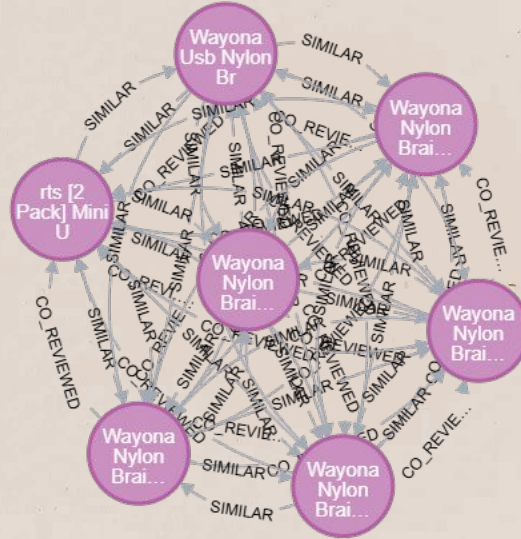
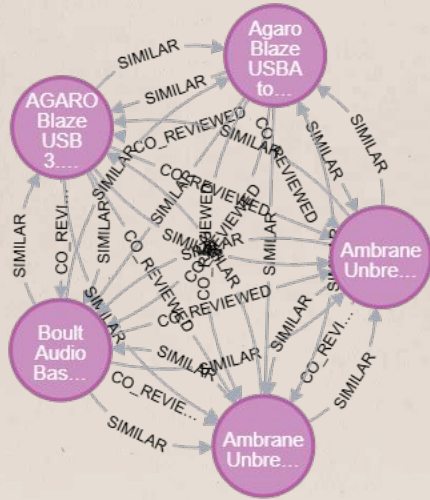


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Additional Visualization



MongoDB Use Cases



Use Case: Product Sentiment Analysis for Amazon Reviews (Positive, Neutral, Negative)

Benefits:

- Schema flexibility, easy storage in JSON
- Easy integration with downstream python models
- Handles semi-structured data well, especially review data as texts
- Good for storage of review data and event logging

Weakness:

- Cannot support rich graph-like relationship, limited discovery among product relationships and recommendation chains

Redis Use Cases



Use Case: Real-time Product Insights, such as "Top currently viewed discounted products"

Benefits:

- Fast retrieval
- Real-time performance
- Quick live dashboard insights

Weakness:

- Memory limited by RAM
- Not ideal for long history insights such as "Most popular product in the past 5 years"
- Cannot handle complex graphic relationships

Summary



The graph-based approach revealed product relationships. The graph algorithms uncovered important products, product groups, and similar items. This enables recommendations, cross-selling, and bundling of products.

Thank You!

